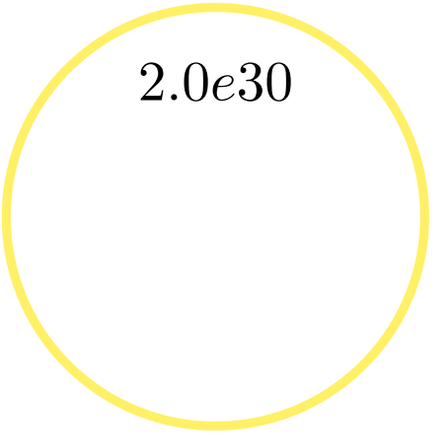
Sun $(0, 1.5e11)$

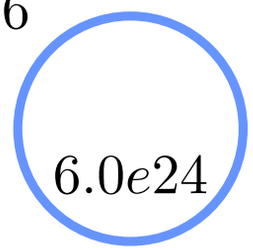


$G = 6.7e-11$

$$\frac{GM_M E_M}{M_D M_D} = \frac{6.7e{-}11 \cdot 7.4e22 \cdot 6.0e24}{3.8e8 \cdot 3.8e8} = 2.1e20\ [m/s^2]$$

$$\frac{GM_M}{M_D M_D} = \frac{6.7e{-}11 \cdot 7.4e22}{3.8e8 \cdot 3.8e8} = 3.4e{-}5\ [kg\ m/s^2]$$

6.4e6



Earth $(0, 0)$

7.4e22

Moon $(3.8e8, 0)$